AIG^{2}AN: Ambiguous-Interpretation Generalized GAN for Self-Supervised Raster-Vector Semantic Segmentation for Cross-Modal Remote Sensing Image

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***Abstract*—Remote sensing image interpretation faces easily overlooked issues: 1) ambiguity: mutual occlusion between objects, e.g., buildings are occluded by trees and vice versa, which leads to ambiguity (building or tree) in GIS, and 2) expression conflicts: topology conflicts between the occluding and occluded objects. However, there have been no reports on the remote sensing ambiguous interpretation (RSAI) or its dataset (RSAID). To address these issues, an RSAID was built, and an ambiguous-interpretation generalized generative adversarial network (AIG^{2}AN) with self-supervised raster-vector semantic segmentation for remote sensing images is proposed for the first time. With the occluding image generated, AIG^{2}AN not only recognizes occlusion regarded as cheating but also restores the occluded semantic labels, which includes three modules: 1) random occlusion generator automatically generates the occluding image with the occluding and occluded labels for self-supervised learning with avoiding hallucination issue; 2) ambiguity dual-interpreters, including both occluding and occluded interpretations, is the first to be proposed; and 3) consistency contrastive learning module (C^{2}LM) is to process the inconsistent interpretations in unambiguous region. Furthermore, we propose a novel multispectral cross-modal learning paradigm via self-supervised pseudo-labeling to automatically enhance the generalization of our AIG^{2}AN in different sensor modalities, and propose a conflict-free and regularized raster-vector expression for interpretation results of ambiguous objects. Experiments on our manually restored RSAID from ISPRS Vaihingen and Potsdam show that AIG^{2}AN improves mIoU and building IoU by 3.28% and 1.80%, with an ambiguity elimination index (AEI) of 85.32% and 76.41%, respectively. Moreover, we reveal the occlusion restoration mechanism. RSAID and code are available at: https://github.com/wenlailiu/AIG2AN**

***Index Terms*—Ambiguous-interpretation generalized generative adversarial network (AIG^{2}AN), conflict-free raster-vector integration, multispectral cross-modal learning paradigm, remote sensing ambiguous interpretation dataset (RSAID), semantic segmentation.**

I. Introduction

REMOTE sensing image interpretation is a vital method for environmental monitoring, land use, and land cover classification [1], change detection [2], [3], etc. Recently, with the increasing complexity of ground objects, mutual occlusions between objects have become more frequent, causing the failure to interpret occluded objects as a significant limitation in applications such as building extraction. During interpretation, the ambiguous phenomenon described above can be observed in Fig. 1. E.g., buildings can be occluded by trees, and the occlusion area is both trees and buildings as a kind of ambiguous object. In building extraction, these ambiguous objects are often interpreted as trees rather than buildings, causing the incompleteness of buildings. This derives from the ambiguity problems, about how the display of ambiguous objects (e.g., buildings and trees, rivers and bridges) overlaps with each other in GIS mapping.

The ambiguity problems cause the decline of segmentation accuracy from remote sensing images and vector topology error in GIS mapping. The former is caused by feature confusion among the ambiguous objects and incomplete interpretation results. The latter occurs due to the loss of key points and lines during the vectorization of occluded objects, which disrupts their original topological structure. For example, buildings occluded by trees lose key edges and corners, rivers occluded by bridges lose connectivity, and occluded roads lose both integrity and connectivity. Therefore, this study defines the ambiguity as an ill-posed issue caused by the projection of 3-D objects onto 2-D image, where occluded features result in missing semantics, disrupted connectivity, and topological inconsistencies.

Fig. 2 illustrates two types of label and topology conflicts: one resulting from the overlapping of interpretation results between occluding and occluded objects in ambiguous regions, and the other caused by inconsistent interpretations in

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Fig. 1. Ambiguous phenomenon: the blue and green lines represent the boundaries of buildings and trees, respectively, and the orange area represents the ambiguous objects of buildings occluded by trees.

unambiguous regions. To address the conflicts, it is essential to develop a novel method for remote sensing ambiguous interpretation (RSAI) and expression. RSAI is simultaneously interpreting occluding and occluded ambiguous objects as well as all other category objects, followed by expressing two interpretation results without conflicts in ambiguous and unambiguous regions.

Currently, common occlusion recognition methods in remote sensing images include cloud detection and occlusion detection. Additionally, there are primarily four methods for occluded restoration. Spatial interpolation and statistical models are simple and feasible, but they are only effective for small areas. Machine-learning methods are widely applied in occluded building and road extraction, but require manual feature construction. Deep-learning methods, such as masked autoencoders (MAEs) [4], generative adversarial networks (GANs) [5], and masked image modeling [6], are used to restore occluded pixels. Recent advancements in MAE-based models (e.g., Alliance [7], GA-MAE [8], and MSA [9]) and diffusion-based methods (e.g., DiffIR [10]) have achieved promising results in image inpainting and representation learning. While these studies have not considered transfer learning in cross-source and cross-temporal scenarios, RSRNet [11] and MARNet [12] have achieved good results in multisource and multitemporal remote sensing interpretation, providing inspiration. However, simultaneous occluding recognition and occluded restoration have not been reported. Consequently, the following directions are worthy of further development.

1) *Simultaneous Ambiguous Interpretation:* There have been no reports on simultaneously interpreting the occluding label and restoring the occluded label. Furthermore, there have been no reports on methods for RSAI or its dataset (RSAID), which includes remote sensing semantic segmentation datasets and occluded labels of all images containing ambiguous objects.
2) *Self-Supervised Occluding and Occluded Sample Generation:* The limited number of ambiguous objects in a remote sensing image is insufficient for network training. Therefore, generating realistic occluding and occluded samples, which follow the distribution patterns of ground objects and avoid the hallucination problem, needs to be explored for self-supervised learning.
3) *Multitask Deep Learning:* Occluding recognition and occluded restoration are not typically supported by conventional GANs. Therefore, a new GAN paradigm is urgently needed for RSAI.
4) *Ambiguous Spatial Expression:* Semantic or topological conflicts arise when expressing RSAI results as raster-vector data. Therefore, exploring how to integrate them compatibly without topology errors is essential.
5) *Cross-Modal Transfer Learning and Generalization:* The models trained on one dataset often suffer from limited generalization to other modalities from other sensors and regions, due to significant domain shifts. Cross-modal learning without labeling is worthy of exploration.
6) *Occluded Restoration Mechanism:* The mechanism for occluded restoration requires further explanation.

To address the aforementioned issues, a remote sensing ambiguity interpretation dataset (RSAID) was constructed, and an ambiguous-interpretation generalized GAN (AIG^{2}AN) with a conflict-free raster-vector integration is proposed for intelligent RSAI and its expression. The major innovations and contributions are summarized as follows.

1) The RSAID we constructed manually restored the labels of occluded objects from the ISPRS Vaihingen and Potsdam semantic segmentation dataset for the first time, providing a foundational dataset for RSAI.
2) The proposed AIG^{2}AN is a novel GAN framework that includes the following key components.
   a) *Random Occlusion Generator:* We propose a new random-walk-masking strategy to enhance Pix2Pix for generating ambiguous samples. Unlike Pix2Pix, which generates the entire image, our strategy ensures that the position, shape, and size of the occlusion image conform to the laws of ground objects. Additionally, it generates more realistic occluding images with their occluding and occluded labels, supporting self-supervised learning for RSAI.
   b) *Ambiguity Dual-Interpreters:* We propose the generalized discriminators of dual-branch networks combined with occluding interpretation and occluded restoration for the first time. Unlike current single-branch networks or discriminators, our discriminators not only segment the occluding image but also recognize the occlusion and restore its occluded label via self-supervised training and joint loss.
   c) *Consistency Contrastive Learning Module (C^{2}LM):* Given the inconsistent results of occluding interpretation and occluded restoration in unambiguous regions, no existing work has addressed this issue. Therefore, we propose a C^{2}LM with supervised learning for the first time. We introduce consistency loss and focal loss to enforce consistency and avoid consistent but false predictions for optimizing interpretation accuracy in unambiguous regions.

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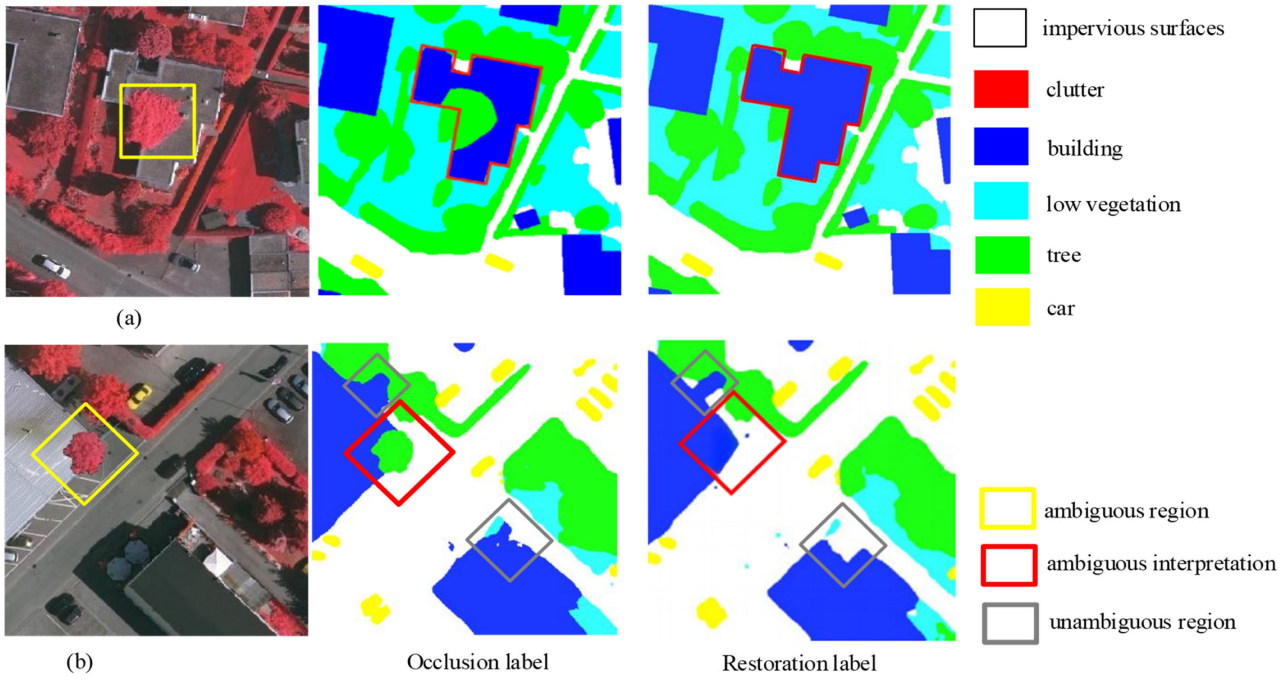


Fig. 2. Ambiguous interpretation of AIG^{2}AN for source (a) image and (b) generated image.

3) *Multispectral Cross-Modal Learning Paradigm:* To the best of our knowledge, we are the first to propose this paradigm via self-supervised pseudo-labeling to enable the generalization of our AIG^{2}AN from one modality (e.g., standard false color images) to other modalities and datasets (e.g., true color images and other false color images), while automatically generating pseudo labels. As a self-supervised training framework, it accomplishes domain adaptation without the need for manual annotations, thereby extending the generalization of our AIG^{2}AN to multispectral cross-modal datasets.
4) *Conflict-Free Raster-Vector Integration:* This integration is proposed for the first time to express raster-vector data with occluding and occluded interpretations in "one map." After vectorizing the raster labels after semantic segmentation, we propose a vector classification optimization strategy, i.e., the polyline compression algorithm is applied to buildings to obtain regularized building outlines, while the Douglas–Peucker algorithm is applied to natural objects, which can improve the segmentation accuracy and simplify the vector expression. Topology errors and label conflicts are resolved by arbitrarily switching the expression of ambiguous regions. Additionally, the consistent expression of unambiguous regions, free from conflict or redundancy, helps save storage space and improves expression efficiency.
5) *Explanation for Occluded Restoration Mechanism:* To our best knowledge, it is the first time to reveal the mechanism of semantic restoration in RSAI.

II. Related Work

A. Semantic Segmentation of Remote Sensing Image

CNNs and transformers, etc., have become the mainstream for semantic segmentation of remote sensing image, with models such as UNet, UNet++, DeepLabv3 [13], EfficientFormerV2 [14], SeaFormer [15], SegFormer [16], ST-Unet [17], UNetFormer [18], and FGNet [19]. The current research on semantic segmentation has developed to focus on the foundation model (SAM [20]) and new architectures (Mamba [21]). The aforementioned networks provide valuable inspiration for our work. We select some of them as the backbones for our ambiguous interpretation.

B. Occluding Recognition of Remote Sensing Image

To address occlusions in remote sensing images, several researchers proposed methods for occlusion recognition and occluded restoration. Lightweight cloud detection networks, such as ECDNet [22], have been proposed.

Different from the aforementioned studies on detecting occlusions, we propose the ambiguity dual-interpreters that jointly occlude interpretation and occluded restoration network for the first time. The networks can simultaneously recognize occlusions and restore the occluded semantic labels.

C. Occluded Restoration of Remote Sensing Image

The methods for occlusion restoration in remote sensing images mainly focus on image processing to restore occluded pixels and labels, e.g., SMDCNet [23] and IPC-Dehaze [24] for cloud removal and dehazing, OARENet [25], SWCARE [26], and MBRE-Net [27] for road extraction, and BIENet [28], MSHFormer [29], and EGAFNet [30] for building extraction.

Unlike the above networks supervised training, ambiguity dual-interpreters are self-supervised trained with generator. Besides, different from typical unsupervised contrastive learning, our consistency contrastive learning is supervised trained with ground truth to ensure consistency and especially enhance interpretation accuracy in unambiguous regions.

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D. Generative Adversarial Network

Currently, GANs have been continuously explored and optimized, e.g., CycleGANs [31], Pix2Pix [32], etc. Conducted semantic embedding GAN [33] can generate photo-realistic remote sensing images by using the semantic mask.

Unlike these GANs, including one generator that learns to produce realistic synthetic data and one discriminator that strives to distinguish between real and generated samples, our AIG2AN designs one generator and two discriminators, which are introduced as follows: Generator, named random-walk-masking Pix2Pix, can generate illusionless occluding images with their occluding labels and occluded labels, while two discriminators can simultaneously interpret the occluding labels and restore the occluded labels, rather than recognizing the authenticity of the generated as other discriminators do.

E. Conflict-Free Raster-Vector Integration

Topology errors and their correction are fundamental issues in GIS mapping that have been progressively addressed. However, this issue has not received sufficient attention in remote sensing mapping and is rarely reported in the literature [34]. This is primarily because remote sensing typically focuses on raster data, often overlooking the topology conflicts that arise when vectorizing raster data.

To address the issues of vector conflicts and raster-vector integration, this article proposes a conflict-free and regularized raster-vector integration method to express the relationships of occlusion and topology between ambiguous objects. When vectorizing interpreted semantic labels, natural objects are simplified by the Douglas–Peucker algorithm, while regular buildings are regularized using the polyline compression algorithm. The unambiguous region is stored only once to optimize storage space.

In summary, the RSAID has been built for deep learning training, and the AIG^{2}AN framework is proposed to achieve: 1) the generation of illusionless occluding images with their occluding and occluded labels for self-supervised learning; 2) simultaneous occluding recognition and occluded restoration; 3) the generalization of our AIG^{2}AN from the ISPRS Vaihingen to the ISPRS Potsdam dataset for RSAI, and 4) conflict-free raster-vector integration without conflict or redundancy.

III. Methodology

To address the ambiguity problems, a new remote sensing ambiguity interpretation dataset (RSAID) has been created, and the AIG^{2}AN is leveraged for RSAI, as shown in Fig. 3. The AIG^{2}AN consists of four modules: the random occlusion generator (G), the ambiguity dual-interpreters, including the occluding interpretation discriminator (OID) network (D_1), occluded restoration discriminator (ORD) network (D_2), C^{2}LM, and conflict-free raster-vector integration.

A. RSAID

The ISPRS Vaihingen and Potsdam semantic segmentation dataset contains numerous images where buildings are occluded by trees. To address this, we manually restored the labels of the occluded buildings, thereby creating the RSAID for the first time. Specifically, the process involved sequential steps: vectorization, restoration of building boundaries based on reference maps, labeling, and rasterization. These steps were applied to obtain the restored labels, as shown in Fig. 4, which were then integrated with the original dataset to form the RSAID. The RSAID provides accurate labels for the occluded objects and resolves the feature confusion of ambiguous objects, as well as the irregularities and nonsmoothness of the building boundaries.

B. Random Occlusion Generator

Given the limited number of ambiguous objects in the RSAID and the high cost of manual restoration, a random-walk-masking-based Pix2Pix approach is leveraged to automatically generate occluding images (e.g., trees occlude buildings) with their occluding label and occluded labels for self-supervised learning. The selection mechanism of the occluding and occluded objects follows the spatial distribution laws of realistic ground objects. For example, there are mutual occlusions of trees and buildings objectively. The specific process is as follows. First, a library of occluding label samples, including images and labels of tree samples with appropriate shape and size, is constructed from the original dataset (as shown in Fig. 3). Second, selected samples from this library randomly walk along the edges of occluded objects (i.e., buildings) to locate the occluding region.

Moreover, the pixels of the nonoccluded region are masked to keep the original image and its labels unchanged, which not only avoids generating nonexistent ground occlusion phenomenon (i.e., hallucinatory images) by the generator but also minimizes unnecessary generation processing. Since the occluding region occupies a relatively small area in each image, the processing speed of generation is improved.

Finally, for the randomly occluding region, the occluding object image $G(\mathbf{X}, \mathbf{Z})$ is generated by Pix2Pix in Fig. 5, using the occluding object label $\mathbf{X}$ and noise $\mathbf{Z}$ as inputs, and then the source image is masked to obtain the occluding image. From the above, it is evident that the entire process of generating both the occluding and occluded objects is self-supervised, well-defined, and controllable, allowing for the simultaneous generation of the occluding image with its occluding and occluded labels.

The loss function of G consists of adversarial loss, pixel loss, and occlusion loss. To ensure that the occluding object image $G(\mathbf{X}, \mathbf{Z})$ generated by G is interpreted as occluding objects by D_1 as effectively as possible, the adversarial loss is expressed as

$$\mathcal{L}_{\text{adv}} = -\mathbb{E}_{\mathbf{X},\mathbf{Z}}\left[\log D(G(\mathbf{X}, \mathbf{Z}), \mathbf{X}, \mathbf{Z})\right] \tag{1}$$

where D represents the discriminator function. To ensure that the pixels of $G(\mathbf{X}, \mathbf{Z})$ are as consistent as possible with the ground truth $\mathbf{Y}$ of the occluding object image, the pixel loss is expressed as

$$\mathcal{L}_{\text{pix}} = \mathbb{E}_{\mathbf{X},\mathbf{Y},\mathbf{Z}}\left[\|\mathbf{Y} - G(\mathbf{X}, \mathbf{Z})\|_1\right]. \tag{2}$$

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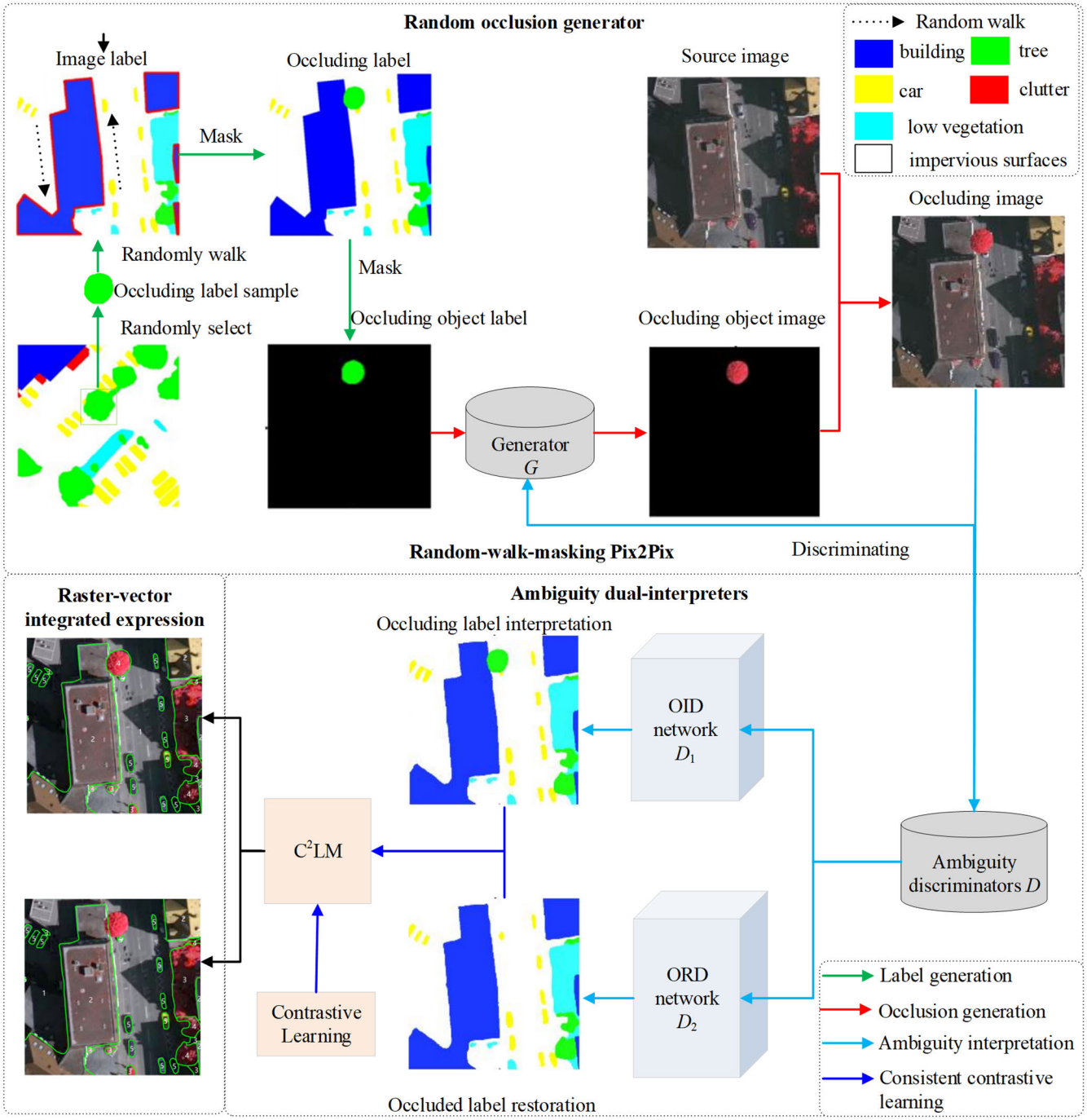


Fig. 3. Architecture of the proposed AIG²AN. The generator, random-walk-masking-based Pix2Pix, generates the occluding image with the occluding label and occluded label. The ambiguity dual-interpreters include the OID network D_1 and the ORD network D_2. C²LM makes the two interpretation results of the unambiguous region consistent by contrastive learning.

To ensure that the occluding image is interpreted as accurately as possible by D_1 without considering occlusion, the occlusion loss is expressed as

$$\mathcal{L}_{\mathrm{CE}} = -\frac{1}{n}\sum_{i=1}^{n}\sum_{c=1}^{m} p_G(x_{ic}) \log q_G(x_{ic}) \tag{3}$$

where n and m represent the total number of samples and categories, respectively; i and c represent the serial number of sample and category for pixel x_i, $p_G(x_{ic})$ denotes the true distribution of labels for occluding interpretation, i.e., when the ground-truth label of x_i is c, $p_G(x_{ic}) = 1$; otherwise, $p_G(x_{ic}) = 0$, and $q_G(x_{ic})$ denotes the probability distribution of labels predicted by D_1.

C. Ambiguity Dual-Interpreters

In G²AN, besides the aforementioned generator G obtaining the occluding image and its occluding labels and occluded labels, there is also a need for adversarizing, i.e., recognizing and interpreting these labels. Thus, dual-branch networks are

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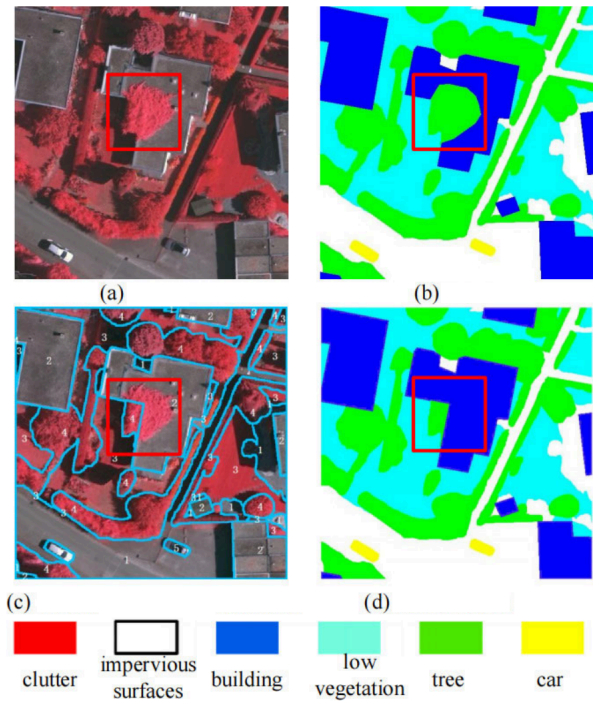


Fig. 4. Manual restoration of ambiguous area: Red frame. (a) Image. (b) Label. (c) Building boundary restoration. (d) Rasterization.

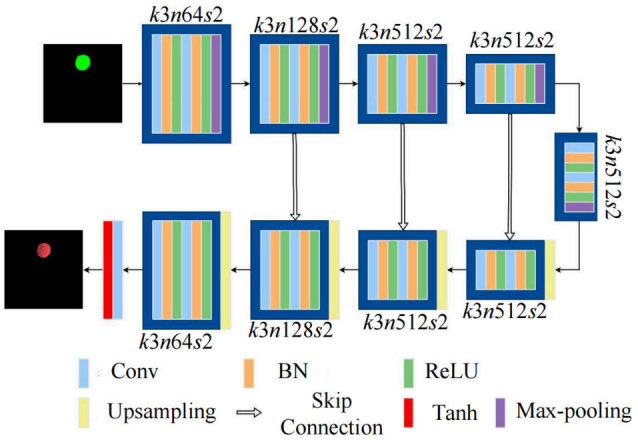


Fig. 5. Network structure of random occlusion generator G [32]. The k, n, and s represent the convolution kernel size, kernel number, and stride, respectively.

trained and tested on the RSAID, with any semantic segmentation network as a backbone. Discriminators are tricked by G, which generates nearly realistic images, but interpret occluding labels and restore occluded labels by D_1 and D_2, respectively. Therefore, D_1 and D_2 are trained to improve the authenticity of occluding images and enhance the accuracy and generalization ability of simultaneously ambiguous interpretation by self-supervised learning.

D_1 does not consider occlusion and directly interprets, i.e., semantic segmentation. An initial model of D_1 is trained on the original dataset, and then D_1 is iteratively trained on the occluding images and their labels generated by the generator.

D_2 automatically recognizes the ambiguous areas and restores their occluded labels. Thus, D_2 is initialized from the model of D_1 and fine-tuned on the restoration dataset from RSAID, followed by iterative training on the occluding images and source image labels as the occluded labels. MAE randomly masks a large portion of the image for self-supervised training and restores the pixel values of the masked regions. Similarly, G randomly generates occluding objects to mask the source image, while D_2 develops the ability to recognize the occluded areas and restore the labels using the context features from the nonoccluded regions through self-supervised training. Our discriminator, D_2, similar to the encoder-decoder structure of MAE, uses a transformer and employs self-attention and cross-attention in the decoding module to capture the spatial context of the image, thereby restoring the semantic labels of the occluded objects.

The loss function of the ambiguity dual-interpreters: The predictions of D_1 should be consistent with the occluding label, so the cross-entropy loss function for D_1 is expressed as

$$\mathcal{L}_{D1} = -\frac{1}{n}\sum_{i=1}^{n}\sum_{c=1}^{m} p_D(x_{ic}) \log q_D(x_{ic}) \tag{4}$$

where n and m represent the total number of samples and categories, respectively; i and c represent the serial number of the sample and category for pixel x_i, $p_D(x_{ic})$ denotes the true distribution of labels for occluding interpretation, and $q_D(x_{ic})$ denotes the probability distribution of labels predicted by D_1.

The predictions of D_2 need to be consistent with the ground truth of occluded restoration, so the cross-entropy loss of D_2 can be expressed as

$$\mathcal{L}_{\mathrm{CE2}} = -\frac{1}{n}\sum_{i=1}^{n}\sum_{c=1}^{m} p^{\tau}(x_{ic}) \log q^{\tau}(x_{ic}) \tag{5}$$

where n and m represent the total number of samples and categories, respectively; i and c represent the serial number of sample and category for pixel x_i, $p^{\tau}(x_{ic})$ denotes the true distribution of labels for occluded restoration, and $q^{\tau}(x_{ic})$ denotes the probability distribution of labels as predicted by D_2.

The Dice loss of D_2 is used to measure the similarity between the segmentation results in the occluded restoration region and the corresponding ground truth. It is defined as

$$\mathcal{L}_{\mathrm{Dice}} = 1 - \mathrm{Dice} = 1 - \frac{2\,|\mathbf{x} \cap \mathbf{x}'|}{|\mathbf{x}| + |\mathbf{x}'|} \tag{6}$$

where $\mathbf{x}$ and $\mathbf{x}'$ represent the ambiguity regions $\mathbf{A}$ of ground truth and differing results predicted by D_1 and D_2, respectively.

The total loss function of D_2 is expressed as

$$\mathcal{L}_{D2} = \mathcal{L}_{\mathrm{CE2}} + \mu\mathcal{L}_{\mathrm{Dice}} \tag{7}$$

where μ is a hyperparameter that balances the two loss terms.

D. Consistency Contrastive Learning Module

Considering that the dual-branch discriminator networks D_1 and D_2 learn distinct features from different datasets, inconsistencies may occur in their interpretation results within the unambiguous region $\mathbf{U}$, outside the ambiguous region $\mathbf{A}$. However, $\mathbf{A}$ can only be accurately identified when the results in $\mathbf{U}$ are consistent; otherwise, discrepancies in $\mathbf{U}$ may be mistakenly recognized as part of $\mathbf{A}$, reducing the accuracy

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of ambiguous region interpretation, e.g., trees not occluding buildings may be misclassified as buildings.

Inspired by the voting-like strategy in ensemble learning, we note that regions with inconsistent predictions are more likely to be misclassified and should receive higher weights in the loss function to guide correction and improve accuracy. In contrast, consistent regions are more likely to be correctly interpreted.

To enforce consistency between the interpretation results L_{D1} and L_{D2} in $\mathbf{U}$ from D_1 to D_2, we propose the C²LM to minimize the inconsistency I between L_{D1} and L_{D2}. Different from existing contrastive learning, C²LM is a supervised learning module that not only enhances the consistency of D_1 and D_2 segmentation outputs L_{D1} and L_{D2} in $\mathbf{U}$, but also ensures they are consistent with ground truth labels. Considering the unreliability of semantic segmentation predictions during initial training, we conduct the contrastive learning of unambiguous regions after the predictions tend to stabilize. Inconsistency I is defined by Iverson Bracket, i.e., Iver$\{\cdot\}$, which means if $\{\cdot\}$ is true, the value is 1, otherwise it is 0, and loss function $\mathcal{L}_{D12}$ can be expressed as

$$I_{ij} = \text{Iver}\left\{L_{D1}(i,j) \neq L_{D2}(i,j)\right\}((i,j) \in \mathbf{U}) \tag{8}$$

$$\mathcal{L}_{D12} = \frac{1}{MN}\sum_{i=1}^{M}\sum_{j=1}^{N} I_{ij} \tag{9}$$

where i and j are the number of pixel rows and columns, M and N are the total number of rows and columns of $\mathbf{U}$, and I_{ij} is the inconsistency between the labels $L_{D1}(i, j)$ and $L_{D2}(i, j)$ interpreted by D_1 and D_2 for the pixel (i, j) in $\mathbf{U}$.

Besides, to control the conflict of the above inconsistent results in $\mathbf{U}$ and align them with ground truth, focal loss [35] $\mathcal{L}_{\text{FL}}$ is used in D_2 to enhance the weight of inconsistent sample loss in inconsistent region $\mathbf{U_{Inc}}$ of $\mathbf{U}$ and leverage ground truth for penalizing false predictions, which can be expressed as

$$\begin{aligned}\mathcal{L}_{\text{FL}} = -\frac{1}{KL}\sum_{i=1}^{K}\sum_{j=1}^{L}\sum_{(i,j)\in\mathbf{U}_{\text{Inc}}} &(\alpha y_{ij}(1-p_{ij})^{\gamma}\log(p_{ij}) \\ &+ (1-\alpha)(1-y_{ij})p_{ij}^{\gamma}\log(1-p_{ij}))\end{aligned} \tag{10}$$

where i and j denote the row and column indices of a pixel, K and L represent the total number of rows and columns in $\mathbf{U_{Inc}}$ with inconsistent conflict, y_{ij} indicates the true distribution of labels for occluded restoration, p_{ij} denotes the probability that D_2 predicts correctly, and α and γ are hyperparameters used to control the weight of inconsistent sample loss and to smoothly adjust the rate at which consistent samples are downweighted.

E. Joint Loss Function

The joint loss function of AIG²AN includes two parts: joint loss $\mathcal{L}_G$ for random occlusion generator G, and joint loss $\mathcal{L}_D$ for ambiguity dual-interpreters consisting of $\mathcal{L}_{D1}$ for the occluding interpretation network D_1, $\mathcal{L}_{D2}$ for the occluded restoration network D_2, $\mathcal{L}_{D12}$ and $\mathcal{L}_{\text{FL}}$ for consistency contrastive learning

$$\mathcal{L}_G = \mathcal{L}_{\text{adv}} + \lambda_1\mathcal{L}_{\text{pix}} + \lambda_2\mathcal{L}_{\text{CE}} \tag{11}$$

$$\mathcal{L}_D = \mathcal{L}_{D1} + \gamma_1\mathcal{L}_{D2} + \gamma_2\mathcal{L}_{D12} + \gamma_3\mathcal{L}_{\text{FL}} \tag{12}$$

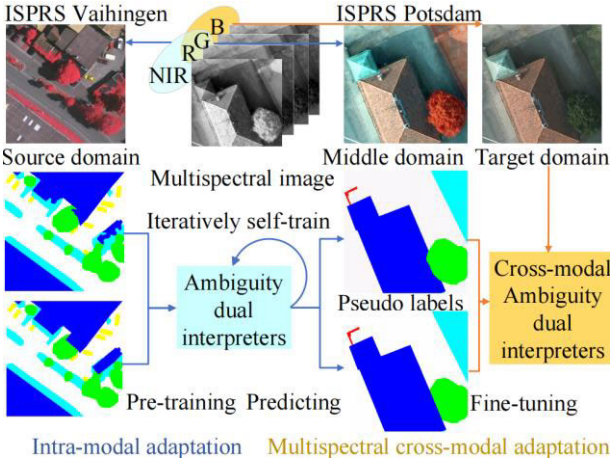


Fig. 6. Architecture of multispectral cross-modal learning paradigm via self-supervised pseudo-labeling.

where λ_1 and λ_2 are the hyperparameters used to balance the three components of $\mathcal{L}_G$ computed by $\mathcal{L}_{\text{adv}}$, $\mathcal{L}_{\text{pix}}$, and $\mathcal{L}_{\text{CE}}$ from (1)–(3); γ_1, γ_2, and γ_3 are the hyperparameters used to balance the four components of $\mathcal{L}_D$ computed by $\mathcal{L}_{D1}$, $\mathcal{L}_{D2}$, $\mathcal{L}_{D12}$, and $\mathcal{L}_{\text{FL}}$ from (4), (7), (9), and (10). These hyperparameters are defined as learnable standard deviations by the uncertainty-weighted loss [36].

F. Multispectral Cross-Modal Learning Paradigm

A multispectral cross-modal learning paradigm via self-supervised pseudo-labeling enables models trained on a source domain with one band combination to adapt to a target domain with a different band combination. For example, in Fig. 6, ambiguity dual-interpreters pretrained on the false-color (NIR, R, G) Vaihingen dataset can be transferred in a self-supervised manner to the true-color (R, G, B) Potsdam dataset.

The core of this paradigm lies in constructing a middle domain within the target multispectral image, which shares the same band combination as the source domain. Using the pretrained model, high-confidence pseudo labels are generated from this middle-domain image. These labels, combined with the inherent spatial alignment of multispectral data, then guide supervised adaptation to the target band composition. The process consists of two stages.

1) Intramodal Spatial Adaptation: The source-domain model, with high accuracy, performs inference on the middle-domain image to produce initial pseudo labels. Only predictions with SoftMax scores above a class-adaptive threshold are retained. This threshold is dynamically adjusted [37] based on the IoU of each class from the source domain, encouraging more samples from well-learned categories to participate. The model is self-trained iteratively for progressively adapting to the middle domain in an "easy-to-hard" manner by gradually converting pseudo labels for all classes into high-quality, trusted labels, thereby enhancing the overall reliability of the ambiguous interpretation.

The quantity $\sigma_e(k, \mathbf{X})$ reflects the learning effect of class k at training step e, calculated as the number of samples x_n

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predicted as class k with a confidence $c_e(x_n)$ of pseudo labels above a fixed threshold τ as below

$$\sigma_e(k, \mathbf{X}) = \sum_{n=1}^{N} \mathrm{Iver}\{\max(c_e(x_n) > \tau)\} \cdot \mathrm{Iver}\left\{\arg\max(c_e(x_n) = k)\right\} \tag{13}$$

where N is the total number of unlabeled patches in the batch, $x_n \in \mathbf{X}$, and $\mathbf{X}$ is the unlabeled dataset, and τ is a predefined fixed threshold ($\tau = 0.96$). A warm-up function $\beta_e(k, \mathbf{X})$ is formulated to normalize this count and better initialize the parameters by

$$\beta_e(k, \mathbf{X}) = \frac{\sigma_e(k, \mathbf{X})}{\max\left\{\max_k\left(\sigma_e, N - \sum_k \sigma_e\right)\right\}}. \tag{14}$$

Finally, a flexible, class-specific threshold $T_e(k, \mathbf{X})$ is updated for each class at step e by scaling the base threshold τ with the warm-up function

$$T_e(k, \mathbf{X}) = \beta_e(k, \mathbf{X}) \cdot \tau^{\mathrm{IoU}^k_{\mathrm{source}}} \tag{15}$$

where $\mathrm{IoU}^k_{\mathrm{source}}$ is the IoU of class k in the source domain. This method ensures that classes with a stronger learning signal (higher σ_e) have a higher effective threshold, allowing more pseudo labels for that class to be included, thereby mitigating sample imbalance.

2) Multispectral Cross-Modal Adaptation: Using the above high-quality pseudo labels as supervision, the self-trained ambiguity dual-interpreters are fine-tuned on the target band set (e.g., R, G, B) to obtain the cross-modal ambiguity dual-interpreters. Instead of full fine-tuning or partial freezing, we employ low-rank adaptation [38] for parameter-efficient updates, minimizing overfitting while enabling effective cross-modal self-supervision.

G. Conflict-Free Raster-Vector Integration

To express pairs of occluding and occluded semantic images with their corresponding vectorized results in one map, a raster-vector integration method is leveraged for the expression of ambiguous objects in Fig. 7. This method includes two key steps.

1) *Classification Vectorization of Results:* Natural objects are simplified by the Douglas–Peucker algorithm to extract contours closer to their shape. Artificial buildings are individually regularized, i.e., leveraging the polyline compression algorithm [39] to make the edge lines of the buildings align to preferred directions and close to orthogonal intersection at the corner point.
2) *Raster-Vector Integration:* As shown in Fig. 7, the raster-vector result interpreted by D_1 and D_2 is $\mathbf{S_{occ}}$ and $\mathbf{S_{res}}$. Therefore, the raster-vector result $\mathbf{U}$ of the unambiguous region, the raster-vector map $\mathbf{O_{occ}}$ of the occluding object, the restored raster-vector map $\mathbf{O_{res}}$ of the occluded object, and the ambiguous region $\mathbf{A}$ containing conflicts can be expressed as

$$\mathbf{U} = \mathbf{S}_{\mathrm{occ}} \cap \mathbf{S}_{\mathrm{res}} \tag{16}$$

$$\mathbf{O}_{\mathrm{occ}} = \mathbf{S}_{\mathrm{occ}} - \mathbf{U} \tag{17}$$

$$\mathbf{O}_{\mathrm{res}} = \mathbf{S}_{\mathrm{res}} - \mathbf{U} \tag{18}$$

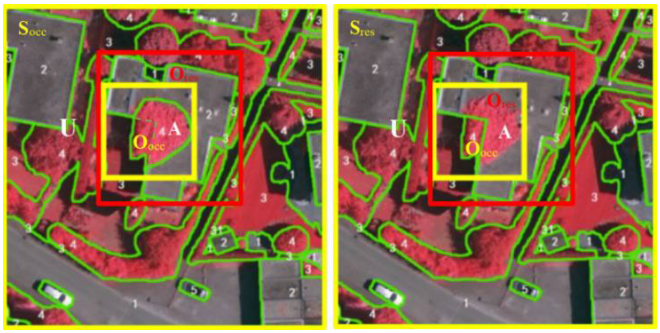


Fig. 7. Conflict-free raster-vector integration of ambiguous objects: occluding objects (trees) $\mathbf{O_{occ}}$ as foreground display in left image, occluded objects (buildings) $\mathbf{O_{res}}$ as foreground display in right image. The green line is the vectorization result, and the white labels 1, 2, 3, 4, and 5, respectively, represent impervious surfaces, buildings, low vegetation, trees, and cars.

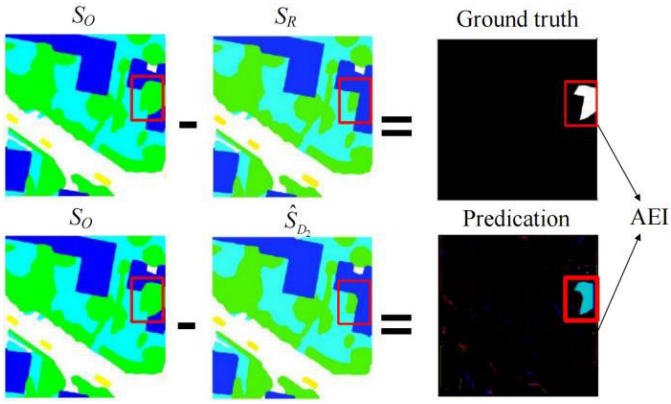


Fig. 8. AEI calculation process. The white area is the ground truth of the ambiguity region, while the cyan area is the prediction of the occluded restoration region by D_2.

$$\mathbf{A} = \mathbf{O}_{\mathrm{occ}} \bar{\cap} \mathbf{O}_{\mathrm{res}}. \tag{19}$$

The raster-vector integrated data $(\mathbf{R}|\mathbf{V})_{\mathrm{int}}$ is defined as

$$(\mathbf{R}|\mathbf{V})_{\mathrm{int}} = \mathbf{U} \cup \mathbf{O}_{\mathrm{occ}} \cup \mathbf{O}_{\mathrm{res}} \tag{20}$$

where $\bar{\cap}$ indicates obtaining the intersecting region.

H. Evaluation Indicator of Ambiguity Interpretation

Traditional semantic segmentation metrics fail to adequately reflect the restoration performance of models for RSAI. To address this, an ambiguity elimination index (AEI) is used to quantify the proportion of the correctly restored area of occluded objects, as predicted by the occluded restoration network D_2, within the ground-truth ambiguity region in Fig. 8. It is defined as

$$\mathrm{AEI} = \frac{\left|(S_O - S_R) \cap (S_O - \hat{S}_{D_2})\right|}{|S_O - S_R|} \tag{21}$$

where S_{O} and S_{R} denote the ground truth in the original and restored datasets, respectively, and $\hat{S}_{D2}$ is the occluded restoration label predicted by D_2. The operation "–" denotes the difference set, i.e., $S_0 - S_R = S_0 - (S_0 \bigcap S_R)$.

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IV. Experiments and Results

To validate the effectiveness of our AIG²AN framework, we design the experiments as follows.

1) *Generator Experiment:* Self-supervised generation of occluding images with occluding labels and occluded labels by random occlusion generator.
2) *Dual-Interpreters Experiment:* Occluding interpretation and occluded restoration by generalized discriminators.
3) *Comparison Experiments of Baselines and Several Backbones, Including Classic and SOTA Networks:* Leveraging image inpainting models, i.e., first MAE [4] or DiffIR [10] to restore the occluded image, and last semantic segmentation networks with the same backbones of AIG²AN to segment the restored image.
4) *Ablation Experiment:* Evaluating performance and module effectiveness with AEI.
5) *Generalization Experiment:* Testing AIG²AN trained by our multispectral cross-modal learning paradigm on ISPRS Potsdam RSAID.
6) *Raster-Vector Integration Experiment:* Providing conflict-free and regularized raster-vector expression results. The specific baseline is designed as follows.

MAE-segmentation and DiffIR-segmentation as baseline models consist of first MAE or DiffIR for occluded image inpainting and last semantic segmentation network for occluded interpretation compared with AIG²AN, respectively. These baseline models incorporate three strategies.

1) *Training With Restored Image:* For images with ambiguous objects, we fuse ambiguous regions (defined as the difference between occluding interpretation and occluded restoration labels in RSAID) with adjacent unambiguous building areas to construct building masks. These guide a Pix2Pix model to generate a realistically restored image. The MAE and DiffIR are then fine-tuned to reconstruct these restored images from their masked versions.
2) *Training With Simulated Occluding Images:* For images without ambiguous objects, we simulate occlusions by randomly walking tree samples along building contours. The MAE and DiffIR are fine-tuned to restore the original buildings from these synthetic inputs.
3) *Determining the Probable Ambiguity Regions:* Based on topological relationships (e.g., adjacency, containment) between trees and buildings, we generate masking buffers along relevant boundaries. The trained MAE and DiffIR subsequently restore content within these regions, respectively.

Finally, semantic segmentation networks interpret the restored images. These three strategies enable MAE-segmentation and DiffIR-segmentation to handle both real and simulated occlusion restoration and interpretation.

The specific experiments and results are as follows.

A. Experimental Settings

1) *Implementation Details:* All experiments were conducted using PaddlePaddle. Training images were cropped via sliding windows with 512 × 512 pixels. The training optimizer for all models was AdamW with a polynomial decay schedule (initial learning rate = 0.0003, the power of the polynomial = 0.9). The batch size was set to 4. We employed focal loss with hyperparameters $\alpha = 0.3$ and $\gamma = 2$. Model performance was compared during convergence. Experiments were performed on a local machine equipped with two NVIDIA GeForce RTX 3060 GPUs (12 GB GPU×2).
2) *Evaluation Metrics:* Typical evaluation metrics for semantic segmentation in remote sensing interpretation include overall accuracy indicators such as MIoU, OA, Kappa, and Dice, as well as category-specific accuracy metrics such as IoU, Precision, and Recall. In addition, the AEI is used to quantify the degree of ambiguity restoration.

B. Results of Random Occlusion Generator

The realistic occluding images with their occluding labels and occluded labels were automatically generated in any image containing buildings by the generator, which randomly selected label samples from the library. Specifically, the selected label samples could randomly walk around different locations in different buildings to generate realistic occluding trees by random-walk-masking-based Pix2Pix, as shown in Fig. 9. Their size and shape conform to real occlusion scenarios as well as the spatial semantic distribution of ground objects, providing samples and labels for the self-supervised training of the discriminators below.

C. Comparison Results of Dual-Interpreters' Backbones

As described in Section III, two ambiguity dual-interpreters, D_1 and D_2, with seven representative semantic segmentation networks, U-Net, UNet++, DeepLabv3 based on CNN, and EfficientFormerV2-L, SeaFormer-L, SegFormer-B5, UNetFormer (with Swin transformer encoder) based on transformer, were evaluated by the generalized adversarial training with the above generators. Additionally, we constructed MAE-segmentation and DiffIR-segmentation as baselines models with U-Net and UNetFormer for comparison with D_2. Representative visual results are presented in Fig. 10, with full quantitative metrics provided in Table I. The results of seven selected networks and baselines are presented below.

For the occluding interpretation task, AIG²AN basic models based on SegFormer-B5 and UNetFormer demonstrate higher accuracy in segmenting various objects, including trees that occlude buildings in Fig. 10. For the occluded restoration task, compared with MAE + UNetFormer and DiffIR + UNetFormer baseline models, occluded restoration networks based on UNetFormer more effectively restore occluded buildings, except for the key points of building corners in the third image (as shown in Fig. 10), due to the difficulty in extracting and restoring contextual features at corners compared to edges. Besides, baseline models with more model parameters for the two networks requires 5–10 h longer time for training than D_2. Furthermore, compared with MAE-segmentation and DiffIR-segmentation, the AEI of UNetFormer basic models increases by 30.43% and 26.69%, indicating that D_2 of AIG²AN more

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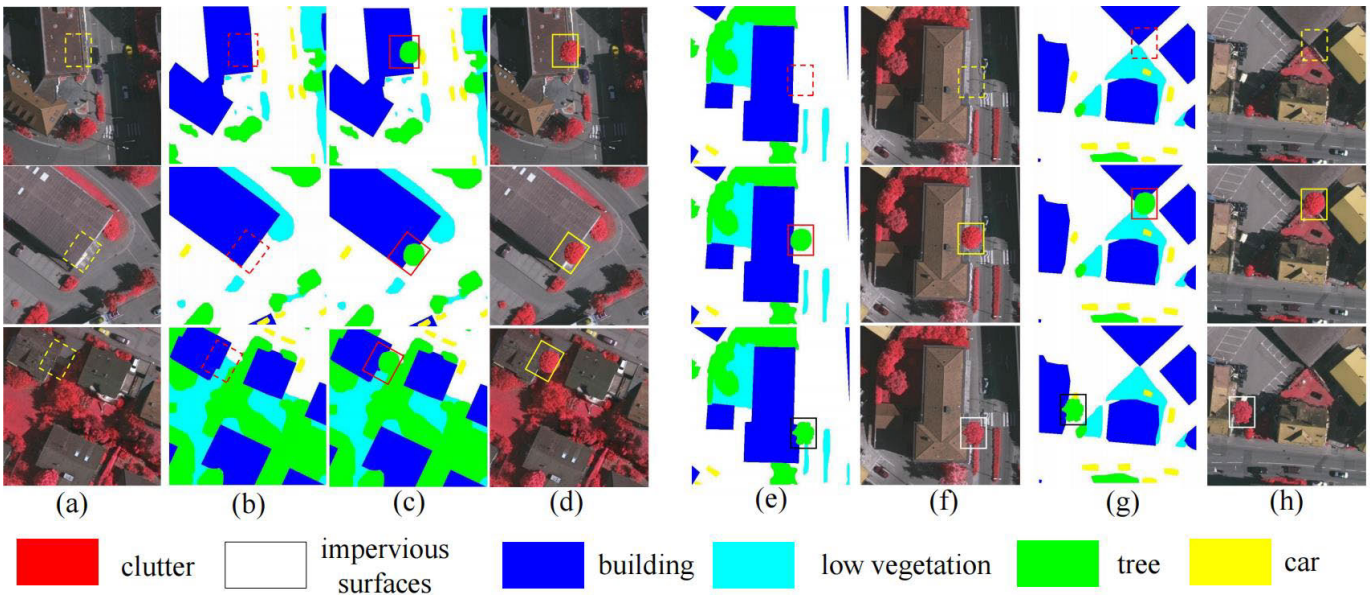


Fig. 9. Results of the random occlusion generator: The same occluding objects (yellow boxes for image and red for its corresponding label) in different images. (a) Source imagery. (b) Occluded labels. (c) Occluding labels. (d) Occluding images. Different occluding objects (black and white boxes) in the same image: (e) and (g) consist of occluded labels and occluding labels. (f) and (h) are occluding images. Solid and dashed boxes represent the original and generated image or label, respectively.

TABLE I

QUANTITATIVE COMPARISON FOR OCCLUDING INTERPRETATION (D_1) AND OCCLUDED RESTORATION (D_2) TASKS OF DIFFERENT MODELS (BASELINES: MAE/DIFFIR-SEGMENTATION AND AIG^{2}AN WITH SEVERAL BACKBONES) ON ISPRS VAIHINGEN RSAID

Model	MIoU(%)		OA(%)		Kappa(%)		Dice(%)		Training time/h		AEI(%)
	D_1	D_2	D_1	D_2	D_1	D_2	D_1	D_2	D_1	D_2	D_2
MAE+U-Net	--	71.48	--	85.92	--	82.73	--	85.29	--	27.62	47.73
MAE+UNetFormer	--	81.37	--	90.36	--	88.47	--	89.16	--	30.65	54.89
DiffIR+U-Net	--	73.34	--	86.51	--	83.02	--	85.82	--	28.73	49.28
DiffIR+UNetFormer	--	81.95	--	91.21	--	88.79	--	90.06	--	32.18	58.63
U-Net	72.25	76.50	87.28	87.69	83.16	83.65	83.47	86.14	21.08	22.50	60.23
UNet++	75.83	78.10	88.60	88.41	84.92	84.60	85.93	87.22	22.12	23.20	65.47
EfficientFormerV2-L	77.46	78.35	89.01	88.60	85.46	84.86	86.97	87.38	**9.82**	**9.95**	68.91
SeaFormer-L	78.69	78.63	89.62	88.81	86.27	85.14	87.77	87.56	10.42	10.75	71.83
DeepLabv3	79.02	78.97	89.84	89.03	86.56	85.42	87.96	87.76	23.75	23.90	75.68
SegFormer-B5	82.67	82.41	92.37	91.78	89.62	89.28	90.37	90.17	23.07	23.33	81.70
UNetFormer	**87.95**	**87.55**	**94.34**	**93.13**	**93.93**	**92.65**	**94.33**	**93.01**	21.56	22.03	**85.32**

effectively restores the semantic labels of occluded buildings. Comparing D_1 and D_2, AIG^{2}AN basic models yield consistent interpretations in unambiguous regions, enabling accurate identification of ambiguous regions. Among them, the UNetFormer basic model achieves superior segmentation of buildings and vehicles in unambiguous regions, exhibiting minimal fragmentation or omission.

Among the seven network models and baselines, the UNetFormer basic model achieves superior segmentation in Fig. 10 and demonstrates the highest accuracy across all items in Table I. However, its four overall accuracy metrics for occluded restoration, MIoU, OA, Kappa, and Dice, decrease by 0.40%, 1.21%, 1.28%, and 1.32%, respectively, compared to those for occluding interpretation. Mutual occlusion between objects causes feature confusion, making "perspective" inference of occluded objects more challenging than direct interpretation, which explains the slightly lower accuracy of D_2 compared to D_1. As shown in Table II, the UNetFormer basic model achieves the highest accuracy across all metrics for buildings, followed by impervious surfaces

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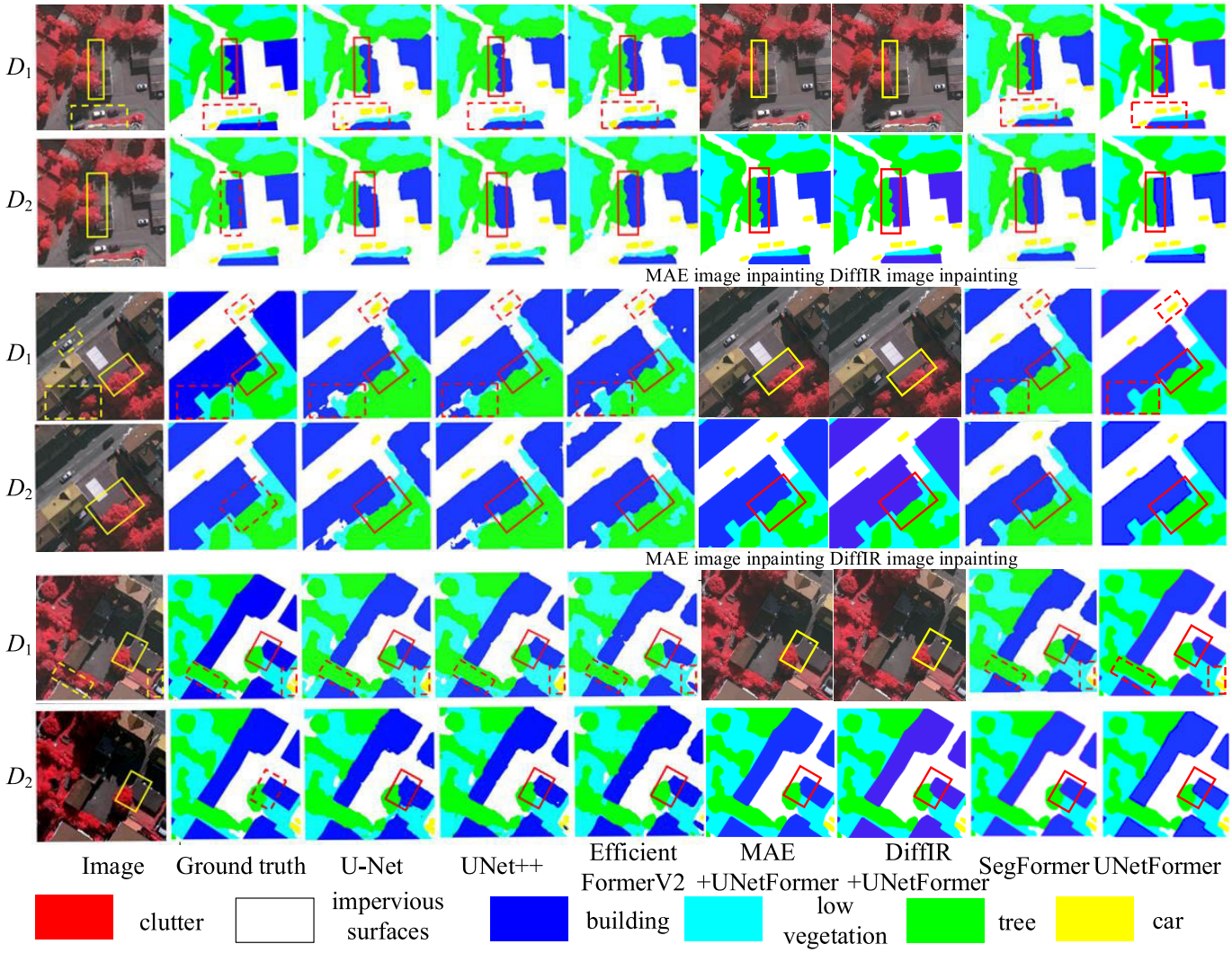


Fig. 10. Qualitative comparison of different models in occluding interpretation and occluded restoration: D_1 for occlusion labels and D_2 for the occluded restoration labels and inpainting images of occluded buildings based on MAE and DiffIR. The yellow solid frame is the ambiguous phenomenon appearing in the image, and the red solid frame is the ambiguous region. The dashed frame is the region where the interpretation may have errors.

TABLE II
ACCURACY FOR OCCLUDING INTERPRETATION (D_1) AND OCCLUDED RESTORATION (D_2) OF UNETFORMER ON ISPRS VAIHINGEN RSAID

Type	IoU(%)		Precision(%)		Recall(%)		AEI(%)
	D_1	D_2	D_1	D_2	D_1	D_2	D_2
0 clutter	84.33	83.41	87.94	86.82	89.67	87.37	--
1 impervious surfaces	91.97	91.82	94.05	93.43	93.32	92.01	--
2 building	**94.12**	**93.88**	**96.32**	**95.37**	**96.22**	**95.14**	85.32
3 low vegetation	82.71	82.21	87.74	84.12	85.45	84.78	--
4 tree	88.14	87.79	90.60	90.39	92.66	92.12	85.32
5 car	86.42	86.21	88.18	87.89	88.31	86.72	--

and trees. This indicates its strong capability in recognizing buildings and impervious surfaces, and comparatively robust performance for trees, with a high recall rate for recognizing all the trees that occlude buildings.

The MIoU of occluded restoration relative to occluding interpretation fluctuates between −0.89% and 4.25% across all networks, with computation time varying within 1.42 h. For the UNetFormer basic model, the accuracy for buildings and

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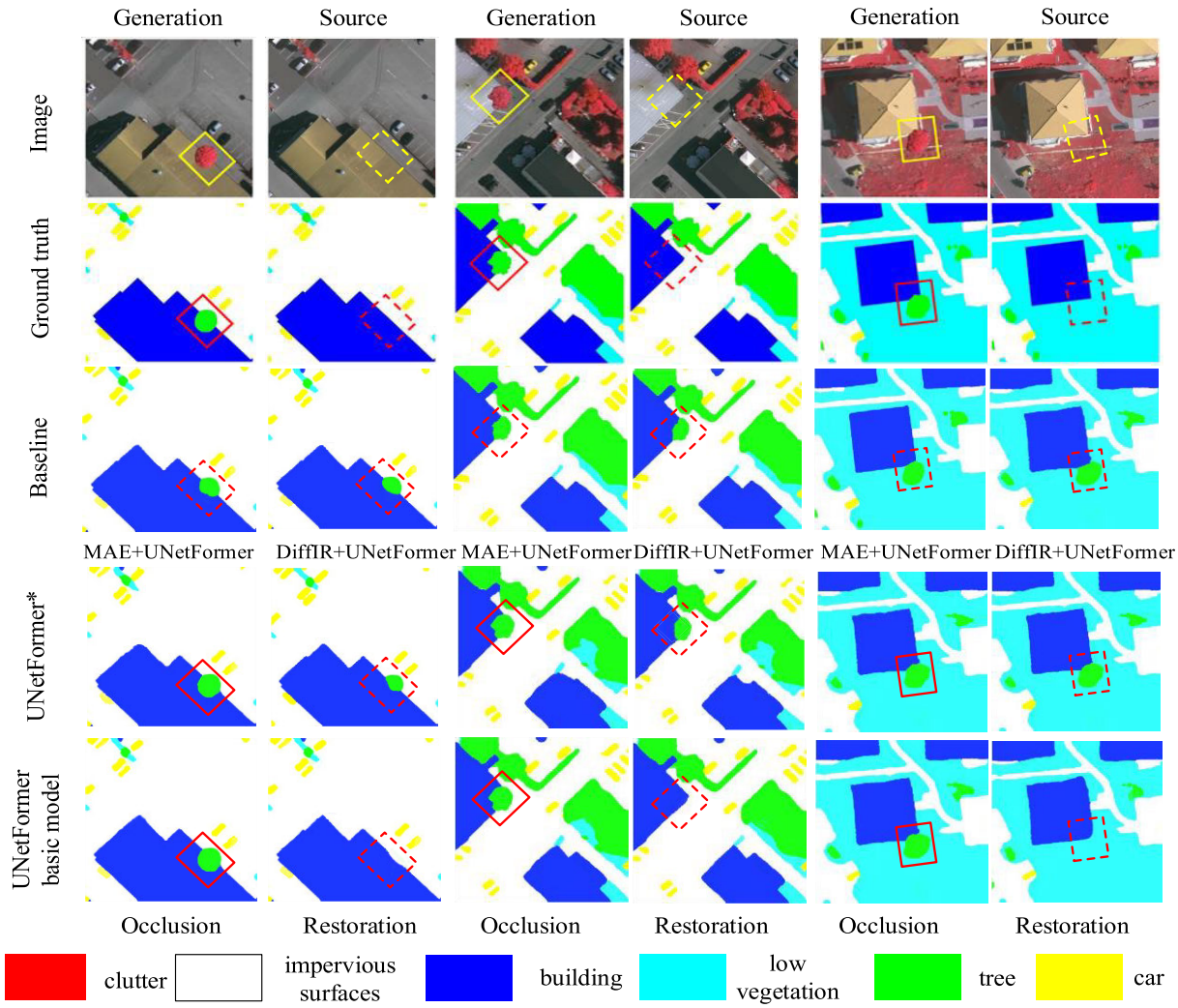


Fig. 11. Comparison among MAE-segmentation and DiffIR-segmentation with UNetFormer, UNetformer* without G^{2}AN and UNetformer basic model with G^{2}AN. Generation is the generated occluding image as input only to be interpreted. Occlusion and restoration are the result of occluding interpretation and occluded restoration. Yellow and red boxes represent the image and the corresponding label of the ambiguity region, respectively. Solid and dashed boxes represent the original and generated image or label, respectively.

trees decreases by 0.24% and 0.35% in Table II, indicating that the occluded restoration network has only a marginal impact on accuracy. The AEI of 85.32% for UNetFormer and 81.70% for SegFormer indicates that D_2 of AIG^{2}AN effectively restores the semantic labels of occluded buildings without a significant accuracy decline.

D. Ablation Study

To evaluate the effectiveness of each module in the AIG^{2}AN framework, ablation experiments were conducted with UNetFormer, which demonstrated superior segmentation performance. Specifically, dual-branch networks based on UNetFormer were directly trained on both RSAID and the realistic dataset generated by the generator, resulting in UNetFormer* without G^{2}AN. This model was tested alongside UNetFormer basic models on the newly generated dataset. For these generated images, we also provided the occluded restoration of the MAE-segmentation and DiffIR-segmentation baseline models. The corresponding segmentation results are presented in Fig. 11.

The basic model achieves finer segmentation of occlusions, demonstrating that the generalized adversarial training of D_1 and G effectively enhances the performance of D_1, and that the samples generated by G are realistic enough to serve as valid occlusion representations. For occluded restoration, although UNetFormer* achieved partial restoration with ground-truth occluded labels, performing on par with the MAE-segmentation and DiffIR-segmentation baselines in Table III, both of them exhibit limited capability to restore finer architectural details, such as building corners and edges. The basic model accurately identified occlusions and more completely restored the occluded buildings, indicating that occluded regions are better restored under the generalized adversarial training of D_2 and G.

For occluding interpretation, the UNetFormer* model achieved MIoU, OA, Kappa, and Dice scores of 84.67%, 91.77%, 88.31%, and 89.84%, respectively. The basic model

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TABLE III
ACCURACY FOR OCCLUDING INTERPRETATION (D_1) AND OCCLUDED RESTORATION (D_2) OF UNETFORMER* WITHOUT G²AN ON ISPRS VAIHINGEN RSAID

Type	IoU(%)		Precision(%)		Recall(%)		AEI(%)
	D_1	D_2	D_1	D_2	D_1	D_2	D_2
0 clutter	79.21	76.19	81.41	79.22	82.45	80.31	--
1 impervious surfaces	90.18	88.37	93.76	92.32	92.85	91.42	--
2 building	**92.32**	**91.03**	**94.13**	**93.29**	**95.01**	**94.13**	62.12
3 low vegetation	80.21	79.91	86.22	84.56	85.28	84.37	--
4 tree	83.22	81.71	86.21	84.51	85.32	82.79	62.12
5 car	82.93	80.59	83.58	82.92	84.41	82.25	--

adversarial training of D_2 and G.

TABLE IV
ABLATION STUDY FOR G²AN AND C²LM OF UNETFORMER

G²AN	C²LM		MIoU(%)		Dice(%)		Consistency(%)	AEI(%)
	$\mathcal{L}_{D12}$	$\mathcal{L}_{FL}$	D_1	D_2	D_1	D_2	in the **U** region	
			84.67	82.97	89.84	89.31	92.41	62.12
√			87.43	86.05	92.89	91.64	93.29	84.89
√	√		87.51	86.77	93.06	91.98	98.34	85.27
√	√	√	**87.95**	**87.55**	**94.33**	**93.01**	**98.79**	85.32

with only G²AN showed improvements of 2.76% and 3.05% for MIoU and Dice, indicating enhanced segmentation performance with G²AN in Table IV. For occluded restoration, UNetFormer* achieved MIoU, OA, Kappa, and Dice scores of 82.97%, 92.87%, 89.77%, and 89.31%, respectively. The basic model showed improvements of 3.08% and 2.33% for MIoU and Dice, also indicating enhanced segmentation performance with G²AN. Additionally, compared with UNetFormer*, the AEI of the basic model with only G²AN has also increased by 22.77% in Table IV, which shows the better restoration of occluded buildings after the generalized adversarial training.

Moreover, to evaluate the effectiveness of the C²LM, additional ablation experiments were conducted based on UNetFormer. As shown in Table IV, the basic model without C²LM produced a 6.71% inconsistency in the unambiguous region between the occluding interpretation and occluded restoration results. With C²LM, this inconsistency was reduced to 1.21%, representing 81.97% reduction. This demonstrates that consistency contrastive learning enables better integration of D_1 and D_2 in interpreting unambiguous regions. Additionally, the MIoU of D_1 and D_2 improved by 0.52% and 1.50%, respectively, indicating that C²LM not only enhances consistency but also improves unambiguous interpretation performance

E. Results of Multispectral Cross-Modal Generalization

To assess the multispectral cross-modal generalization capability of AIG²AN, the UNetFormer basic model pretrained on the ISPRS Vaihingen RSAID (source domain) was adapted to the ISPRS Potsdam RSAID (target domain) via self-supervised pseudo-labeling. The quantitative results are summarized in Table V, with qualitative examples shown in Figs. 12 and 13.

After adaptation, D_1 achieved MIoU, OA, Kappa, and Dice scores of 81.25% (83.53% without clutter), 86.71%, 84.93%, and 85.69%, respectively, while D_2 attained 78.46% (81.08% without clutter), 83.43%, 81.12%, and 82.85%. These results indicate that occluded restoration remains intrinsically more challenging than occluding interpretation under cross-modal conditions. Relative to the source domain, the MIoU reductions of 6.70% (D_1) and 9.09% (D_2) reflect the expected spectral domain shift. Importantly, structured classes (e.g., buildings, impervious surfaces, cars) exhibited substantially smaller accuracy degradation than the average MIoU loss, suggesting that AIG²AN effectively learns transferable geometric-semantic representations across spectral modalities. In contrast, clutter experienced a high accuracy drop, attributable to its extremely small proportion (0.8%) and high intraclass variability in the source domain.

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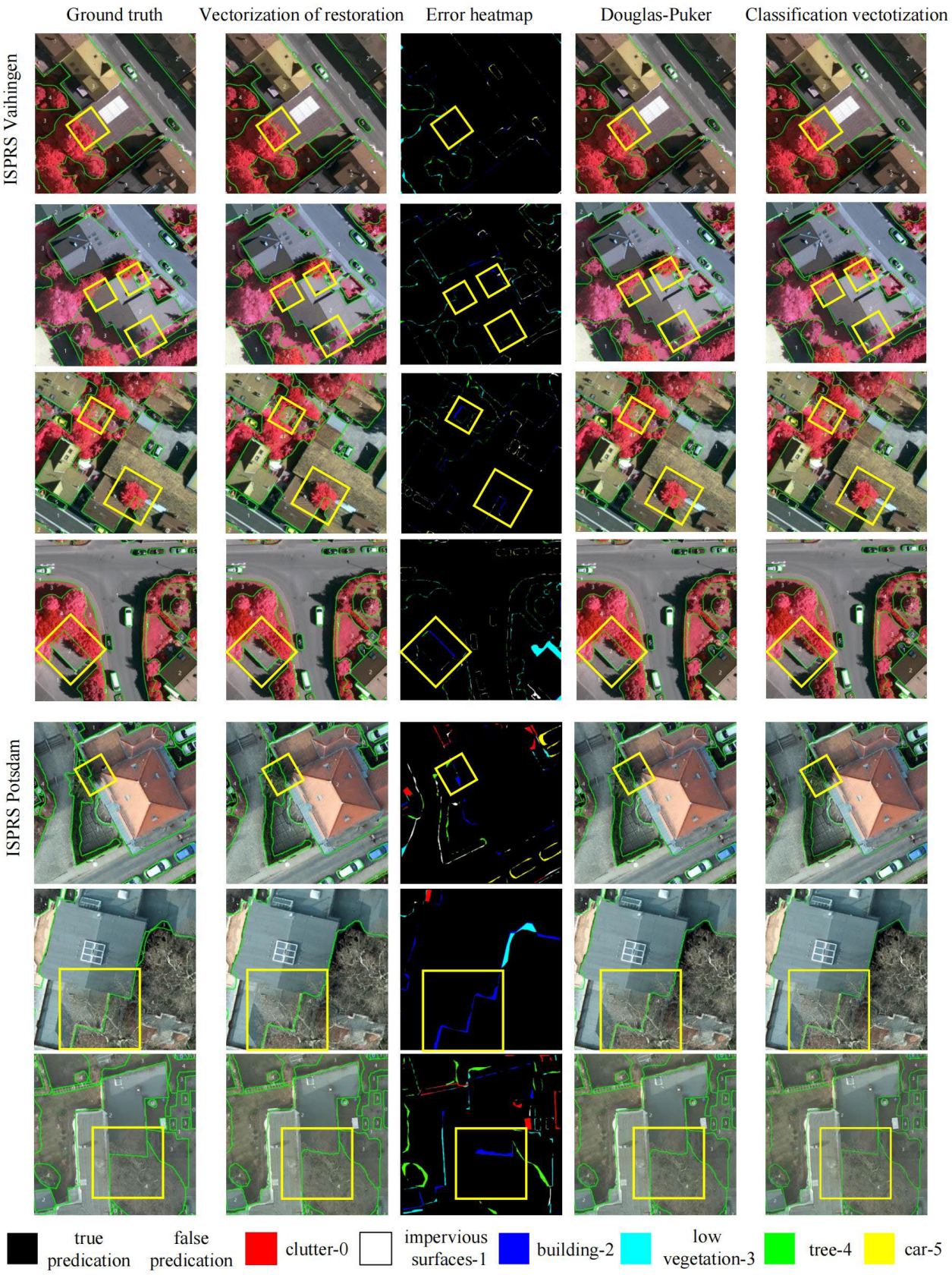


Fig. 12. Multispectral cross-modal generalization experiment with vectorization and error heatmaps. The yellow frame represents the ambiguous objects.

As demonstrated in Figs. 12 and 13, the cross-modal D_2 achieved a near-complete restoration for heavily occluded buildings with an AEI of 76.41%. This illustrates that AIG^{2}AN effectively generalizes across spectral modalities via

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TABLE V
ACCURACY FOR OCCLUDING INTERPRETATION (D_1) AND OCCLUDED RESTORATION (D_2) OF UNETFORMER CROSS-MODAL MODEL ON ISPRS POTSDAM RSAID

Type	IoU(%)		Precision(%)		Recall(%)		AEI(%)
	D_1	D_2	D_1	D_2	D_1	D_2	D_2
0 clutter	69.84	65.31	74.31	73.21	73.56	72.02	--
1 impervious surfaces	88.21	86.11	90.34	88.46	90.89	88.32	--
2 building	**89.32**	**87.45**	**92.31**	**90.29**	**91.47**	**89.45**	76.41
3 low vegetation	77.31	72.25	79.58	76.59	77.60	75.37	--
4 tree	80.35	79.26	84.39	83.23	83.47	82.95	76.41
5 car	82.46	80.35	85.95	83.40	86.62	83.51	--

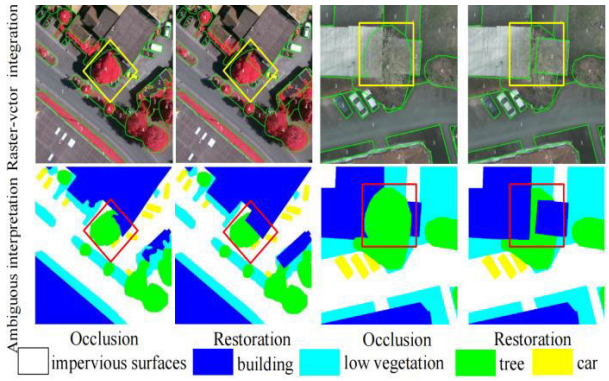


Fig. 13. Conflict-free raster-vector integration results of ambiguous objects. After Douglas simplification and buildings regularization, the results of both expression (yellow) and final ambiguous interpretation (red) are improved.

a multispectral cross-modal learning paradigm. Moreover, in cases where buildings were partially visible under sparse vegetation, the cross-modal D_2 accurately restored the labels of occluded buildings, further validating its strong contextual reasoning and adaptation capacity.

F. Results of Conflict-Free Raster-Vector Integration

Following the ambiguity interpretation, the results from the basic model and the cross-modal model were automatically vectorized, as shown in Fig. 12. Compared with the ground truth of manually restored buildings in Section IV-B, the occluded restoration network effectively restored most of the building areas occluded by trees; however, their straight contours, especially the corners, were not fully restored. To address this, vectorization results were refined using Douglas simplification and building regularization based on the polyline compression algorithm [39] that makes the boundary of buildings close to orthogonal alignment. The former, without separate treatment of the buildings, shows that the outline of the building is smoother, but with many bends and some removed corners. The vector results after building regularization are basically the same as the ground truth, so they are displayed and stored as the final vector results.

The vectorized interpretation results were subsequently expressed through raster-vector integration of ambiguous objects, as illustrated in Fig. 13. Each pair of left and right raster-vector images corresponds to the foreground display of the ambiguous interpretation. Additionally, the unambiguous region exhibited consistent raster-vector results by minimizing the consistency contrastive loss via C²LM. Furthermore, effective interpretation in ambiguous areas was facilitated by the proposed focal loss, which targeted errors arising from inconsistent conflicts.

V. DISCUSSION

A. RSAID

The RSAID restores labels of occluded objects based on the ISPRS Vaihingen and Potsdam dataset; thus, its quality is determined by the effectiveness of occluded restoration. As shown in Fig. 10, Tables I and V, all backbone networks achieve high restoration accuracy, indicating the high quality of RSAID and its suitability as a benchmark dataset for ambiguous interpretation.

B. Random Occlusion Generator

Our generator obtained realistic occlusion images with both occluding and occluded labels, as shown in Fig. 9, adhering to occlusion distribution laws. Unlike [32] and [33], which generate images without semantic labels, our generator simultaneously produces occluding images with their occluding labels and occluded labels, enabling fully self-supervised training. The controlled masking ensures known occlusion locations and ground truth in occluded regions.

Random-walk-masking strategy avoids the hallucination problem in GANs like Pix2Pix, reduces computational costs by only generating the occluding regions, and retains original nonoccluded areas. This improves efficiency, minimizes artifacts, and shortens generation time. Additionally, the occlusion mechanism follows real-world object distribution (e.g., trees occluding buildings at contours), enhancing realism and physical plausibility, particularly with adversarial training.

C. Ambiguity Dual-Interpreters

The ambiguity dual-interpreters D_1 and D_2 outperformed in occluding interpretation and occluded restoration, respectively,

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in Figs. 11 and 12. Inspired by MAE [4], D_2 restored over 85.32% of occluded labels in Table II by leveraging spatial context via self-supervised training on randomly masked images and their original labels. While MAE reconstructs entire images from heavily masked inputs, D_2 focuses on restoring smaller occluded labels, benefiting from richer spatial context. Masked image reconstruction of MAE is a regression task predicting RGB values, while semantic label restoration of D_2 is a classification task with a limited set of labels, making it faster and more efficient. D_2 achieves 87.55% MIoU in just 22.03 h of training, demonstrating significant efficiency gains over MAE.

D. Baselines Comparison of MAE/DiffIR-Segmentation

The MAE-segmentation and DiffIR [10]-segmentation baseline first restored the occluded image and then segmented the restored image, requiring two sequential networks for occluded interpretation. The training of these networks is independent, leading to the accumulation of errors. Additionally, it relies on an extra spatial relationship to identify probable ambiguous regions that may not be true and need manual intervention to determine masks for MAE and DiffIR. Compared with this, our ambiguity dual-interpreters are an end-to-end model that directly obtains occluding and occluded labels without generating intermediate restored images. The lower model parameters and less storage space improve the efficiency, and end-to-end interpretation avoids error accumulation, resulting in higher accuracy. Moreover, our method automatically and clearly identifies ambiguous regions in Fig. 7, without relying on spatial relationships or manual intervention for the mask of MAE and DiffIR.

E. Consistency Contrastive Learning Module

This module improved the interpretation consistency of dual discriminators in unambiguous regions, enhancing accuracy as shown in Table IV. Different from typically unsupervised contrastive learning methods relying on high-dimensional similarity metrics, we use Iverson brackets to compute consistency directly on low-dimensional label maps, addressing the inconsistencies more efficiently. Our contrastive learning is supervised and follows a voting-like strategy in ensemble learning: consistent predictions indicate higher accuracy, while the inconsistent signal indicates potential error. We apply Focal loss [35] to align the inconsistent results with ground truth, thereby avoiding convergence on consistent but erroneous predictions. It not only ensures consistency but also enhances interpretation accuracy in unambiguous regions.

F. Multispectral Cross-Modal Learning Paradigm

This paradigm effectively achieved multispectral cross-modal transfer for RSAI in Fig. 12 and Table V by decomposing the process into manageable steps, introducing a middle domain as a bridge. By employing a dynamic thresholding strategy [37] for pseudo-label selection and a progressive domain transition mechanism, it successfully mitigates the issue of noise accumulation. Furthermore, the cross-modal model fine-tuning strategy is optimized, ultimately achieving robust domain adaptation without the need for manual annotation.

G. Conflict-Free Raster-Vector Integration

The raster-vector results of D_1 and D_2 are split into ambiguous and unambiguous regions, simplifying the expression, as shown in Figs. 12 and 13. Regularization [39] enhances structural rationality and standardization of occluded buildings, improving occluded restoration and reducing space complexity, while the Douglas–Peucker algorithm simplifies irregular objects. In ambiguous regions, topological conflicts are avoided by separately storing occluding and occluded objects as the foreground. In unambiguous regions, consistency loss ensures unified storage and visualization, reducing redundancy and storage overhead.

H. Effectiveness and the Restored Mechanisms

The ablation study demonstrates that the proposed AIG^{2}AN model outperforms the model without G^{2}AN, as evidenced in Fig. 11, Tables II and III. Based on both CNN and transformer backbones, AIG^{2}AN excels in occluded restoration (Fig. 10). Key mechanisms include.

1) *Ground Object Distribution Conformity:* Our random occlusion generator mimics real-world scenarios, e.g., trees occluding buildings, by having "tree " pixels walk along building contours. This ensures that the generated occlusions and restorations are authentic and geometrically consistent with man-made objects
2) *Data-Driven Large Model Alignment:* By leveraging diverse, high-quality datasets, our random occlusion generator creates a wide range of realistic samples, enhancing model accuracy and generalization in ambiguous phenomena.
3) *MAE-Like Restoration Principle:* The network restores masked regions using contextual information, learning regular building features to recognize and reconstruct occluded buildings, enabling effective ambiguous interpretation.
4) *Self-Adaptive Multiscale Token Restoration:* Using Swin transformer in UNetFormer, we apply a multiscale token mechanism to handle varying occlusion sizes, avoiding structural distortions at occlusion boundaries. This enables more accurate restoration of objects like trees, which vary in scale.

I. Limitations

Our proposed AIG^{2}AN focuses on interpreting occluded man-made objects with regular structures (e.g., buildings) and their occluding objects. While it primarily addresses building-tree ambiguities, it can be extended to other regular man-made objects (e.g., rivers occluded by bridges, roads occluded by vehicles), as these exhibit learnable structural patterns and can be manually annotated for evaluation. Adapting to new scenarios involves adjusting the random-walk-masking strategy based on the geometry of target object: 1) for bridges

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occluding rivers, the random-walk trajectory is perpendicular to the river centerline, generating elongated bridge masks aligned transversely across the river channel with their length aligned with the width of river and 2) for vehicles occluding roads, the random walk is constrained within the road polygon with mask shapes biased toward compact rectangles oriented along the road direction.

However, its performance with naturally irregular occluded objects remains untested, primarily due to the challenge of manually annotating high-fidelity ground truth for such objects, which lack the structural regularity of man-made ones. Nevertheless, if occlusion-free images and semantic labels for natural objects become available, AIG^{2}AN should be applicable to these cases.

VI. Conclusion

This study addresses the challenge of ambiguity in remote sensing interpretation. To this end, we constructed the RSAID based on the ISPRS Vaihingen and Potsdam dataset and propose the AIG^{2}AN framework for simultaneous interpretation of ambiguous objects, along with a conflict-free raster-vector integration of results. The main conclusions are as follows.

1) *Accuracy and Effectiveness:* AIG^{2}AN demonstrates high accuracy and scientific validity. With UNetFormer as the backbone, the framework achieves MIoU scores of 87.95% and 87.55% for occluding interpretation and occluded restoration, respectively.
2) *Component Effectiveness:* Each module contributes significantly to performance, including the random occlusion generator with self-supervised learning, the ambiguity dual-interpreters with fine-tuning, and the consistency contrastive learning strategy, all of which enhance interpretation accuracy.
3) *Vector Classification Optimization:* The strategy produces contours better aligned with object boundaries, improves the segmentation accuracy of occluded buildings, and reduces the number of vector nodes, optimizing expression, accuracy, and space complexity.
4) *Raster-Vector Integration:* The integrated raster-vector expression not only eliminates topology conflicts but also substantially reduces data storage requirements.
5) *Cross-Modal Generalization:* The multispectral cross-modal learning paradigm enables AIG^{2}AN to achieve ambiguity interpretation on datasets with different spectral modalities without requiring manual annotations.
6) *Model Flexibility:* The models of G^{2}AN are replaceable, allowing integration with advanced backbone networks. While this study focuses on the new AIG^{2}AN paradigm, comparison with the latest models is reserved for future work.
7) *Paradigm Advancement:* AIG^{2}AN offers a novel framework for remote sensing intelligent interpretation and efficient raster-vector integrated mapping.

Addressing the ambiguity interpretation and raster-vector expression ensures the semantic and topological completeness of extracted geographic information, increasing the reliability of geospatial applications. Future work will involve constructing datasets for naturally irregular occluded types to further expand the applicability of AIG^{2}AN high-accuracy GIS mapping.

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